Matemàtiques B
4t ESO

SOLUCIONS DE LA FITXA · UNITAT 2

Polinomis i fraccions algebraiques

FentMates

Intenta resoldre cada exercici abans de mirar-ne la solució. Pots comprovar les divisions, Ruffini, les arrels, les factoritzacions i les fraccions amb les eines interactives dels apunts de la unitat.

1 Polinomis

1. ★☆☆

a) $\boxed{2x^4+3x^2-x-5}$: grau 4, coeficient principal 2, terme independent -5. No és complet (falta x^3).

b) $\boxed{-x^3+7}$: grau 3, coeficient principal -1, terme independent 7. No és complet.

c) $\boxed{x^5+x^3-2x^2-8x+1}$: grau 5, coeficient principal 1, terme independent 1. No és complet (falta x^4).

d) $\boxed{6x^2-4x-\frac{1}{2}}$: grau 2, coeficient principal 6, terme independent $-\frac{1}{2}$. És complet.

2. ★☆☆

a) -3, x^4, $\boxed{4}$

b) $\frac{2}{5}$, xy^2, $\boxed{3}$

c) 8, cap, $\boxed{0}$

d) -1, a^3b^2, $\boxed{5}$

3. ★☆☆

a) $\boxed{2x^2-3x+3}$

b) $\boxed{5x^3+x^2-3x}$

c) $\boxed{x^2-6x}$

2 Operacions amb polinomis

4. ★☆☆

a) $\boxed{2x^3+3x+3}$

b) $\boxed{2x^3-2x^2-3x+5}$

c) $\boxed{x^3+x^2-7x+2}$

d) $\boxed{2x^4-5x^3+2x^2+4x-8}$

e) $\boxed{x^3-2x^2+7x+2}$

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5. ★☆☆

a) $\boxed{x^2+10x+25}$ **b)** $\boxed{4x^2-12x+9}$ **c)** $\boxed{9x^2-1}$

d) $\boxed{x^4-4x^2+4}$ **e)** $\boxed{\frac{1}{4}x^2+4x+16}$ **f)** $\boxed{-x^2+25}$

6. ★★☆

a) $\boxed{(x+3)^2}$ **b)** $\boxed{(2x-5)^2}$ **c)** $\boxed{(x+7)(x-7)}$ **d)** $\boxed{(3x+1)(3x-1)}$

7. ★★☆

a) $\boxed{8x}$ **b)** $\boxed{-x^2-6x+12}$ **c)** $\boxed{2x^2-4x+3}$

8. ★★☆

a) $c(x)=\boxed{x-2}$, $r(x)=\boxed{2x+1}$

b) $c(x)=\boxed{3x^2-2}$, $r(x)=\boxed{3x+2}$

c) $c(x)=\boxed{2x^2+3x+1}$, $r(x)=\boxed{-7x+1}$

d) $c(x)=\boxed{x^2-1}$, $r(x)=\boxed{0}$

9. ★☆☆

a) $c(x)=\boxed{x^2-2x-3}$, $r=\boxed{0}$

b) $c(x)=\boxed{2x^3-2x^2-x+2}$, $r=\boxed{-7}$

c) $c(x)=\boxed{x^4+2x^3+4x^2+8x+16}$, $r=\boxed{0}$

d) $c(x)=\boxed{3x^2-4x+7}$, $r=\boxed{-14}$

10. ★★☆ $(x+a)(2x-3)=2x^2+(2a-3)x-3a$. Igualant coeficients: $-3a=-6\Rightarrow\boxed{a=2}$ i $b=2a-3\Rightarrow\boxed{b=1}$.

3 Valor numèric i teorema del residu

11. ★☆☆

a) $\boxed{-1}$ **b)** $\boxed{-1}$ **c)** $\boxed{-7}$ **d)** $\boxed{-1}$ **e)** $\boxed{-\frac{5}{8}}$

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12. ★★☆

a) $P(1) = 1 - 3 + 2 = \boxed{0}$

b) $P(-1) = -1 - 2 - 3 = \boxed{-6}$

c) $P(2) = 16 - 2 + 5 = \boxed{19}$

d) $P(-1) = 1 - 1 = \boxed{0}$

13. ★★☆ Ruffini amb 2: coeficients $1, 0, -5, 3 \to 1, 2, -1, 1$; el residu és $\boxed{1}$. I $P(2) = 8 - 10 + 3 = 1$. Coincideixen.

14. ★★☆

a) $P(3) = 27 + 9k - 15 + 6 = 0 \Rightarrow \boxed{k = -2}$

b) $P(-2) = -16 + 2k + 4 = 8 \Rightarrow \boxed{k = 10}$

c) $P(-2) = 16 + 4k - 4 = 0 \Rightarrow \boxed{k = -3}$

15. ★★★ $P(2) = 0$ i $P(-3) = 0$: $\begin{cases} 4a + 2b = -2 \\ 9a - 3b = 33 \end{cases} \Rightarrow \boxed{a = 2, b = -5}$.

4 Arrels d'un polinomi

16. ★☆☆

a) $P(1) = 0$: $\boxed{\text{sí}}$ **b)** $P(2) = 0$: $\boxed{\text{sí}}$ **c)** $P(-1) = 12$: $\boxed{\text{no}}$ **d)** $P(-3) = 0$: $\boxed{\text{sí}}$

17. ★★☆

a) $\boxed{x = 1, x = 2, x = 3}$

b) $\boxed{x = 1, x = -1, x = -2}$

c) $\boxed{x = \pm 1, x = \pm 3}$

d) $\boxed{x = 2, x = -2, x = 3}$

18. ★★☆ Divisors de 12: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$. Només $P(2) = 8 + 4 - 12 = 0$: l'única arrel és $\boxed{x = 2}$ (el quocient $x^2 + 3x + 6$ no té arrels reals).

19. ★★☆ $(x-1)(x+2)(x-3) = \boxed{x^3 - 2x^2 - 5x + 6}$; $(x-4)^2 = \boxed{x^2 - 8x + 16}$ (o qualsevol múltiple).

5 Factorització de polinomis

20. ★☆☆

a) $\boxed{3x(x-4)}$

b) $\boxed{x(x+3)(x-3)}$

c) $\boxed{2x(x+2)^2}$

d) $\boxed{x^2(x+1)(x-1)}$

e) $\boxed{5(x+1)(x-1)}$

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21. ★★☆

a) $\boxed{(x-2)(x-3)}$

b) $\boxed{(x+4)(x-3)}$

c) $\boxed{2\left(x+\frac{1}{2}\right)(x-1)=(2x+1)(x-1)}$

d) Irreductible ($\Delta=-4<0$).

e) $\boxed{3(x+3)(x-1)}$

22. ★★☆

a) $\boxed{(x-1)(x+2)(x-3)}$

b) $\boxed{(x-1)^2(x+3)}$

c) $\boxed{(x-1)(x+1)(x+2)(x-3)}$

d) $\boxed{(x+1)^2(x-2)}$

e) $\boxed{(x-1)(x+1)(x-2)(x+2)}$

f) $\boxed{(x-1)^2(x+1)}$

23. ★★★

a) $\boxed{x^2(x-3)(x+1)}$

b) $\boxed{x(x-1)(2x-1)(x+2)}$

c) $\boxed{x(x-1)(x+1)(x^2+1)}$

d) $\boxed{(x-1)(x+1)(x^2+x+1)}$

24. ★☆☆ $x=0$ ($\boxed{\text{doble}}$), $x=3$ ($\boxed{\text{triple}}$), $x=-1$ ($\boxed{\text{simple}}$). Grau $2+3+1=\boxed{6}$.

6 Divisibilitat. M.c.d. i m.c.m.

25. ★☆☆

a) $1-1=0$: $\boxed{\text{sí}}$

b) $4+4=8\neq 0$: $\boxed{\text{no}}$

c) $1-1=0$: $\boxed{\text{sí}}$

d) $9-9=0$: $\boxed{\text{sí}}$

26. ★★☆

a) $(x+2)(x-2)$, $(x+2)^2$: m.c.d. $=\boxed{x+2}$, m.c.m. $=\boxed{(x+2)^2(x-2)}$

b) $x(x-1)$, $(x+1)(x-1)$: m.c.d. $=\boxed{x-1}$, m.c.m. $=\boxed{x(x+1)(x-1)}$

c) $x(x+1)(x-1)$, $(x-1)^2$: m.c.d. $=\boxed{x-1}$, m.c.m. $=\boxed{x(x+1)(x-1)^2}$

d) $2(x+2)$, $x(x+2)$, $3x$: m.c.d. $=\boxed{1}$, m.c.m. $=\boxed{6x(x+2)}$

27. ★★★ $P(x)=(x-1)(x+1)(x-2)$, $Q(x)=(x-1)(x-2)$. m.c.d. $=\boxed{(x-1)(x-2)}$, m.c.m. $=\boxed{(x-1)(x+1)(x-2)}$.

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7 Fraccions algebraiques

28. ★☆☆

a) $\boxed{x = 2}$ **b)** $\boxed{x = \pm 3}$ **c)** Sempre en tenen. **d)** $\boxed{x = 0,\ x = 4}$

29. ★★☆

a) $\boxed{\dfrac{x-3}{x}}$ **b)** $\boxed{\dfrac{x-2}{x+2}}$ **c)** $\boxed{\dfrac{x-1}{2}}$

d) $\boxed{x-1}$ **e)** $\boxed{\dfrac{x-2}{x+2}}$

30. ★★☆

a) $\boxed{\text{Sí}}$: $(x+1)(x^2-1) = (x-1)(x^2+2x+1)$ (o bé, simplificant la segona: $\frac{(x+1)^2}{(x+1)(x-1)} = \frac{x+1}{x-1}$).

b) $\boxed{\text{No}}$: $2(x^2+1) \neq x(2x+2)$.

31. ★★☆

a) $\boxed{\dfrac{2x+3}{x^2}}$ **b)** $\boxed{\dfrac{2x}{x^2-1}}$

c) $\boxed{\dfrac{x^2+4}{x^2-4}}$ **d)** $\boxed{\dfrac{4-x}{x^2-1}}$

e) $\boxed{\dfrac{2x+3}{x(x+1)}}$

32. ★★☆

a) $\boxed{2x(x-2)}$ **b)** $\boxed{\dfrac{1}{x}}$

c) $\boxed{(x-1)(x-3)}$ **d)** $\boxed{\dfrac{1}{2x(x-1)}}$

33. ★★★

a) $\boxed{\dfrac{1}{x-1}}$ **b)** $\boxed{x+1}$

c) $\boxed{1}$ **d)** $\boxed{\dfrac{8x}{x^2-4}}$

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8 Problemes

34. ★★☆ Altura: $(x^2+5x+6):(x+3)=\boxed{x+2 \text{ cm}}$. Perímetre: $2(x+3)+2(x+2)=\boxed{4x+10 \text{ cm}}$.

35. ★★☆ $\dfrac{x^3+6x^2+8x}{x(x+2)}=\dfrac{x(x+2)(x+4)}{x(x+2)}=\boxed{x+4 \text{ cm}}$.

36. ★☆☆ $(x^3-x+5)-(3x^2-2x+1)=\boxed{x^3-3x^2+x+4}$.

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